

Final examination on Algebraic Specification

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1 Binary tree specification

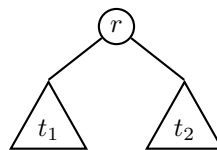
Let us recall an algebraic specification of binary trees over nodes. Let us call it **BIN-TREE**(**node**). Here is the signature:

- **Defined types**

- The type of the binary trees is noted **t**.
- The type of the nodes is noted **node**.

- **Constructors**

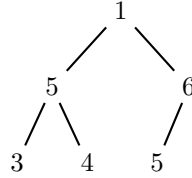
- **EMPTY** : **t**
The term **EMPTY** represents the empty tree, i.e. the tree that contains no node.
- **MAKE** : **node** $\times$ **t** $\times$ **t** $\rightarrow$ **t**
The term **MAKE**(r, t_1, t_2) denotes the tree whose root is r , left subtree is t_1 and right subtree is t_2 . Graphically:



- **Functions**

1. **MEM** : **t** $\times$ **node** $\rightarrow$ **bool**
The term **MEM**(t, e) is **TRUE** if node e occurs in tree t , otherwise it is equal to **FALSE**. (We assume we have equality on nodes.)
2. **MIN-DEPTH** : **t** $\times$ **node** $\rightarrow$ **int**
The term **MIN-DEPTH**(t, e) denotes the minimum depth at which node e occurs in tree t . (The root is at depth 0.) If e is not in t , the value is unspecified. In case the node e appears several times in t , the value is the smallest depth of occurrence. (We assume

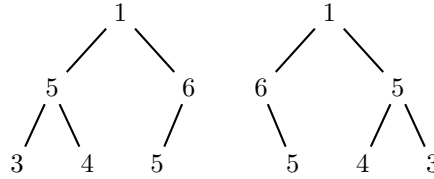
we can compare nodes for equality and that we have the function $\text{MIN} : \text{int} \times \text{int} \rightarrow \text{int}$ which returns the smallest integer argument.) For example, if t is



$$\text{then } \begin{cases} \text{MIN-DEPTH}(t, 7) & \text{is unspecified (i.e. undefined)} \\ \text{MIN-DEPTH}(t, 1) & = 0 \\ \text{MIN-DEPTH}(t, 3) & = 2 \\ \text{MIN-DEPTH}(t, 5) & = \text{MIN}(1, 2) = 1 \end{cases}$$

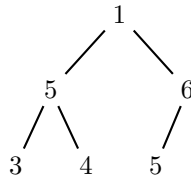
3. $\text{INV} : \mathbf{t} \rightarrow \mathbf{t}$

The term $\text{INV}(t)$ denotes the tree t in a mirror, i.e. the left subtrees of t are the right subtrees of $\text{INV}(t)$ and the right subtrees become the left subtrees. Therefore $\text{INV}(\text{INV}(t)) = t$. For example, the following trees are mirrors of each other:



4. $\text{SUM} : \text{BIN-TREE}(\text{int}).\mathbf{t} \rightarrow \text{int}$

The term $\text{SUM}(t)$ denotes the sum of all (integer) nodes in tree t . If t is empty, the sum is unspecified. For example, if t is



$$\text{then } \text{SUM}(t) = 1 + 5 + 3 + 4 + 6 + 5 = 24.$$

Question 1. Complete this signature with equations defining the functions.

Question 2. Why a breadth-first search is better than a depth-first search in defining MIN-DEPTH?